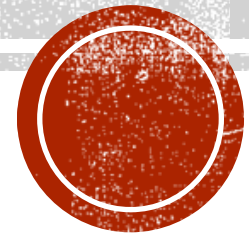


MODULE 4: SEMANTICS



- Computational Desiderata for Representations
- Meaning Structure of Language
- First Order Predicate Calculus
 - Elements of FOPC
 - The Semantics of FOPC

CONTENTS



INTRODUCTION

- Semantics: Meaning for the text
- The text is checked for meaningfulness.
- Eg: Hot ice-cream, Colourless green idea – doesn't give proper meaning.



Semantics Labelling:

“The Keys, which were needed to access the building, were locked in the car”

The Keys, which were needed to access the
building

Argument

Locked
Verb

In the car
Location



ASPECTS OF SEMANTICS ANALYSIS

- **Lexical Processing** – meaning of the word in the sentence
- **Word sense disambiguation** – understanding a word used in which contest
Eg: Bank – Finance or River side
- **Semantics Grammars** – Parsing is done but there is no meaning for the sentence.
- Eg: **John is a Professor at London** – Named Entity Recognition



MEANING REPRESENTATIONS

- We're going to take the same basic approach to meaning that we took to syntax and morphology
- We're going to create representations of linguistic inputs that capture the meanings of those inputs.
- But unlike parse trees, these representations aren't primarily descriptions of the structure of the inputs...



CONTD...

- The frameworks that are used to specify the syntax and semantics of these representations will be called **meaning representation languages**.
- Representations that can bridge the gap from **linguistic inputs** to the kind of **non-linguistic knowledge**
- Needed to perform a variety of tasks involving the meaning of linguistic inputs.



CONTD...

Consider the following everyday language tasks that require some form of semantic processing.

- ✓ Answering an essay question on an exam
- ✓ Deciding what to order at a restaurant by reading a menu
- ✓ Learning to use a new piece of software by reading the manual
- ✓ Realizing that you've been insulted
- ✓ Following a recipe



CONTD...

- For example, some of the knowledge of the word needed to perform the above tasks includes:
- Answering and grading essay questions requires background knowledge about
 - The topic of the question
 - The desired knowledge level of the students
 - How such questions are normally answered



CONTD...

Reading a menu and deciding what to order

- Giving advice about where to go to dinner
- Following a recipe
- Generating new recipes all require deep knowledge about food, its preparation
- What people like to eat and what restaurants are like.



CONTD...

Learning to use a piece of software by reading a manual

- Giving advice about how to do the same
- Requires deep knowledge about current computers
- The specific software in question
- Similar software applications
- Knowledge about users in general



SEMANTIC ANALYSIS

- In the representational approach being explored here,
- The linguistic inputs and construct meaning representations that are made up of the same kind of stuff that is used to represent this kind of everyday commonsense knowledge of the world.
- The process whereby such representations are created and assigned to linguistic inputs is called semantic analysis



Computational Desiderata for Representations

- Representations that allow a system to
 - Answer questions
 - Determine truth
 - Perform inference
- ✓ Verifiability
- ✓ Unambiguous Representations
- ✓ Canonical Form
- ✓ Inference and Variables
- ✓ Expressiveness



VERIFIABILITY

- Verifiability: The system's ability to compare representations to facts in memory.
- The most straightforward way to implement this notion is make it possible for a system to compare, or match, the representation of the meaning of an input against the representations in its knowledge base, its store of information about its world.



EXAMPLE

Does Maharani serve vegetarian food?

`Serves(Maharani; Vegetarian Food)`

- Input - matched against the knowledge base of facts about a set of restaurants.
- matching - the input proposition in its knowledge base, it can return an affirmative answer.
- Otherwise, it must either **say No**, if its knowledge of local restaurants is complete, or **say that it does not know**.



UNAMBIGUOUS REPRESENTATIONS

- The domain of semantics - is subject to ambiguity.
- Single linguistic inputs can legitimately have different meaning representations assigned to them based on the circumstances in which they occur.



CONTD...

- Example

I wanna eat someplace that's close to ICSI

- The answer generated by the system for this request will depend on which interpretation is chosen as the correct one.
- The various linguistic phenomenon that give rise to such ambiguities.



CONTD...

- A concept closely related to ambiguity is vagueness.
- Like ambiguity, vagueness can make it difficult to determine what to do with a particular input based on its meaning representation.
- Vagueness, however, does not give rise to multiple representations.
- Consider the following request as an example.

I want to eat Italian food



CONTD...

- use of the phrase **Italian food** may provide enough information for a restaurant advisor to provide reasonable recommendations.
- It is nevertheless quite vague as to what the user really wants to eat.
- A vague representation of the meaning of this phrase may be appropriate for some purposes, while a more specific representation may be needed for other purposes.



CANONICAL FORM

- The notion that **single sentences can be assigned multiple meanings** leads to the related phenomenon of distinct inputs that should be assigned the same meaning representation.
- Does Maharani have vegetarian dishes?
- Do they have vegetarian food at Maharani?
- Are vegetarian dishes served at Maharani?
- Does Maharani serve vegetarian fare?



CONTD...

- **Canonical form:** inputs that mean the same thing should have the same meaning representations
- Simplifies reasoning
- Makes representations more compact (fewer different representations)
- BUT: makes semantic analysis harder
 - Need to figure out that “have” and “serve” mean the same thing in the previous examples; same for the various phrases for vegetarian food
 - BUT: can perform word sense disambiguation; use a single representation for all senses in a synset



CONTD...

- Consider the issue of the meanings of the words **food, dish and fare** in these examples.
- A little introspection, or a glance at a dictionary, reveals that these words have a fair number of distinct uses.
- These words all have various word senses and that some of the senses are synonymous with one another.
- The process of choosing the right sense in context is called **word sense disambiguation, or word sense tagging** by analogy to part-of-speech tagging.



CONTD...

- Alternative syntactic analyses often have meanings that are, if not identical, at least systematically related to one another.
- Consider the following pair of examples.

Maharani serves vegetarian dishes

Vegetarian dishes are served by Maharani

- use knowledge of where grammatical subjects and direct objects appear in these constructions
- to assign Maharani - to the role of the server
- vegetarian dishes - to the role of thing being served in both of these examples



INFERENCE AND VARIABLES

Can vegetarians eat at Maharani?

- The term inference to refer generically to a system's ability to draw valid conclusions based on the meaning representation of inputs and its store of background knowledge.
- It must be possible for the system to draw conclusions about the truth of propositions that are not explicitly represented in the knowledge base, but are nevertheless logically derivable from the propositions that are present.



CONTD...

I'd like to find a restaurant where I can get vegetarian food.

- In this examples, this request does not make reference to any particular restaurant.
- The user is stating that they would like information about an unknown and unnamed entity that is a restaurant that serves vegetarian food.
- Answering this request requires a more complex kind of matching that involves the use of variables.
- A representation containing such variables as follows.

`Serves(x; Vegetarian Food)`



EXPRESSIVENESS

- A meaning representation scheme must be expressive enough to handle an extremely wide range of subject matter.
- **First Order Predicate Calculus** is expressive enough to handle quite a lot of what needs to be represented.



- <https://www.youtube.com/watch?v=H05PvN-mgtY>



TASK IN SEMANTIC ANALYSIS

Lexical Analysis

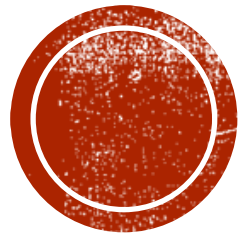
**Relationship
Extraction**

**Named Entity Recognition
(NER)**

**Natural Language
Generation
(NLG)**

**Word Sense
Disambiguation
(WSD)**





MEANING STRUCTURE OF LANGUAGE



CONTD...

- These include a variety of conventional form
 - Meaning associations
 - Word-order regularities
 - Tense systems
 - Conjunctions and quantifiers
 - A fundamental **predicate-argument structure**



CONTD...

- **Predicate-argument structure**, which is the mechanism that has had the greatest practical influence on the nature of meaning representation languages.
- **Events, actions and relationships** can be captured with representations that consist of predicates and arguments to those predicates.
- Languages display a division of labor where some words and constituents (typically) function as predicates and some as arguments.



CONTD...

- Predicates
 - Primarily Verbs, VPs, Sentences
 - Sometimes Nouns and NPs
- Arguments
 - Primarily Nouns, Nominals, NPs, PPs



CONTD...

- These examples can be classified as having one of the **three syntactic argument frames**.

I want Italian food - NP want NP

I want to spend less than five dollars - NP want Inf-VP

I want it to be close by here - NP want NP Inf-VP

- These syntactic frames specify the number, position and syntactic category of the arguments that are expected.



CONTD...

The frame for the variety of want that appears in Example 1 specifies the following facts:

- There are two arguments to this predicate.
- Both arguments must be NPs.
- The first argument is pre-verbal and plays the role of the subject.
- The second argument is post-verbal and plays the role of the direct object



CONTD...

Semantic roles and Semantic restrictions on these roles

- The notion of a semantic role can be understood by looking at the similarities among the arguments in Examples 1 to 4.
- The study of roles associated with specific verbs and across classes of verbs is usually referred to as **thematic role or case role**.
- The notion of **semantic restrictions** arises directly from these semantic roles.



CONTD...

Consider the following phrases from the BERP corpus.

An Italian restaurant under fifteen dollars

- In this example, the meaning representation associated with the preposition *under* can be seen as having something like the following structure.

Under(Italian Restaurant ; \$15)

- Prepositions can be characterized as two-argument predicates where the *first argument is an object* that is being placed in some relation to the *second argument*.



CONTD...

- Another non-verb based predicate-argument structure example.

Make a reservation for this evening for a table for
two persons at 8.

- The predicate-argument structure is based on the concept underlying the noun reservation, rather than make, the main verb in the phrase.
- This example gives rise to a four argument predicate structure like the following.

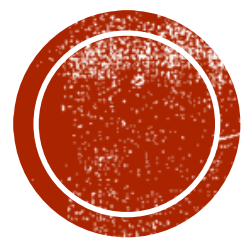
Reservation(Hearer; Today; 8PM ; 2)



CONTD...

- Any useful meaning representation language must be organized - supports the **specification of semantic predicate-argument structures**.
- This support must include support for the kind of semantic information that languages present:
 - **Variable arity predicate-argument structures**.
 - **The semantic labeling of arguments to predicates**.
 - **The statement of semantic constraints on the fillers of argument roles**.





FIRST ORDER PREDICATE CALCULUS



CONTD...

- First Order Predicate Calculus (FOPC) is a flexible, well-understood, and computationally tractable approach to the representation of knowledge that satisfies many of the requirements.
- Terms refer to elements of the domain: **constants, functions, and variables**

Martin, Spouse Of (Karen), X



CONSTANTS

- Constants in FOPC refer to **specific objects** in the world being described.
- Such constants are conventionally depicted - as A and B or single capitalized words that are often indicative of proper nouns such as Maharani and Harry.
- Like programming language constants, FOPC constants refer to exactly one object.



FUNCTIONS

- Functions in FOPC correspond to concepts that which are often expressed in English as genitives such as the **location of Maharani or Maharani's location**.
- A FOPC translation of such an expression
Location Of (Maharani)
- FOPC functions are syntactically the same as single argument predicates.
- Functions provide a convenient way to refer to specific objects without having to associate a named constant with them.
- Restaurants, location associated with them.



VARIABLE

- The notion of a variable is our final FOPC mechanism for referring to objects.
- Variables, which are normally depicted as single lower-case letters, give us the ability to make assertions and **draw inferences about objects without having to make reference** to any particular named object.
- This ability to make statements about anonymous objects comes in two flavors:
 - **making statements about a particular unknown object**
 - **making statements about all the objects in some arbitrary word of objects**



$$\begin{aligned}
 \text{Formula} &\rightarrow \text{AtomicFormula} \\
 &| \text{Formula Connective Formula} \\
 &| \text{Quantifier Variable}, \dots \text{Formula} \\
 &| \neg \text{Formula} \\
 &| (\text{Formula})
 \end{aligned}$$

$$\text{AtomicFormula} \rightarrow \text{Predicate}(\text{Term}, \dots)$$

$$\begin{aligned}
 \text{Term} &\rightarrow \text{Function}(\text{Term}, \dots) \\
 &| \text{Constant} \\
 &| \text{Variable}
 \end{aligned}$$

$$\text{Connective} \rightarrow \wedge \mid \vee \mid \Rightarrow$$

$$\text{Quantifier} \rightarrow \forall \mid \exists$$

$$\text{Constant} \rightarrow A \mid \text{VegetarianFood} \mid \text{Maharani} \dots$$

$$\text{Variable} \rightarrow x \mid y \mid \dots$$

$$\text{Predicate} \rightarrow \text{Serves} \mid \text{Near} \mid \dots$$

$$\text{Function} \rightarrow \text{LocationOf} \mid \text{CuisineOf} \mid \dots$$

Figure 14.2 A context-free grammar specification of the syntax of First Order Predicate Calculus representations. (Adapted from Russell and Norvig (1995).)



FOPC

- FOPC representation for **Maharani serves vegetarian food** might look like the following formula.

Serves(Maharani; Vegetarian Food)

- This FOPC sentence asserts that Serves, a two-place predicate, holds between the objects denoted by the constants Maharani and Vegetarian Food.
- Different use of predicates is illustrated - **Maharani is a restaurant.**
- **Restaurant(Maharani)** - example of a one-place predicate that is used, not to relate multiple objects, but rather to assert a property of a single object.



CONTD...

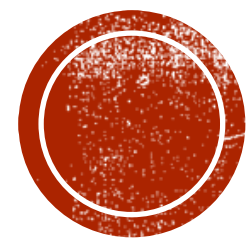
- From Figure 14.2, logical connectives give us the ability to create larger representations by conjoining logical formulas using one of three operators

I only have five dollars and I don't have a lot of time.

$\text{Have}(\text{Speaker}; \text{Five Dollars}) \wedge \neg \text{Have}(\text{Speaker}; \text{Lot Of Time})$

- The semantic representation for this example is built up in a straightforward way from semantics of the individual clauses through the use of the $\wedge$ and $\neg$ operators.





SEMANTICS OF FOPC



CONTD...

- The various objects, properties, and relations represented in a FOPC knowledge base acquire their meanings by virtue of their correspondence to objects, properties, and relations out in the external world being modeled by the knowledge base.
- FOPC sentences - assigned a value of True or False based on whether the propositions they encode are in accord with the world or not.



CONTD...

- Consider the following example.

Ay Caramba is near ICSI.

For this example, such an effort might yield something like the following.

Near(Location of (AyCaramba); Location of (ICSI))

The meaning of this logical formula then arises from the relationship between the terms

Location of (AyCaramba), Location of (ICSI)

The predicate Near, and the objects and relation they correspond to in the word being modeled.



CONTD...

- This sentence can be assigned a **value of True or False**.
- For our current purposes, we will adopt what is known as a database semantics for determining the truth of our logical formulas.
- The interpretations of formulas involving logical connectives is based on the meaning of the components in the formulas combined with the meanings of the connectives they contain.
- Figure gives interpretations for each of the logical operators.



P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$
<i>False</i>	<i>False</i>	<i>True</i>	<i>False</i>	<i>False</i>	<i>True</i>
<i>False</i>	<i>True</i>	<i>True</i>	<i>False</i>	<i>True</i>	<i>True</i>
<i>True</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>True</i>	<i>False</i>
<i>True</i>	<i>True</i>	<i>False</i>	<i>True</i>	<i>True</i>	<i>True</i>

Figure 14.3 Truth table giving the semantics of the various logical connectives.

The semantics of the $\wedge$ (and), and $\neg$ (not) operators
 $\vee$ (or) operator and $\Rightarrow$ (implies) operator



VARIABLES AND QUANTIFIERS

- **variables are used in two ways in FOPC:** to refer to particular anonymous objects and to refer generically to all objects in a collection.
- These two uses are made possible through the **use of operators known as quantifiers**.
- The two operators that are basic to FOPC are the existential quantifier, which is denoted $\exists$, and is pronounced as “there exists”, and the universal quantifier, which is denoted $\forall$, and is pronounced as “for all”.



CONTD...

- Consider the following example.

A restaurant that serves Mexican food near ICSI.

The following would be a reasonable representation of the meaning of such a phrase.

xRestaurant(x)

^Serves(x; Mexican Food)

^Near((Location of (x); Location of (ICSI))



CONTD...

- For example, if AyCaramba is a Mexican restaurant near ICSI, then substituting AyCaramba for x results in the following logical formula.

Restaurant(AyCaramba)

^Serves(AyCaramba; Mexican Food)

^Near((Location Of (AyCaramba); Location Of (ICSI))

- Based on the semantics of the ^ operator, this sentence will be true if all of its three component atomic formulas are true.



CONTD...

- The use of the universal quantifier also has an interpretation based on substitution of known objects for variables.
- The substitution semantics for the universal quantifier takes the expression for all quite literally;
- The $\forall$ operator states that for the logical formula in question to be true the substitution of any object in the knowledge base for the universally quantified variable should result in a true formula.



CONTD...

- Consider the following example.

All vegetarian restaurants serve vegetarian food

- A reasonable representation for this sentence would be something like the following.

$$\forall x \text{VegetarianRestaurant}(x) \Rightarrow \text{Serves}(x, \text{VegetarianFood})$$



CONTD...

- Case where the substituted object actually is a vegetarian restaurant; one such substitution would result in the following sentence.

Vegetarian Restaurant(Maharani)

=> Serves(Maharani; Vegetarian Food)

- WKT the consequent clause,

Serves(Maharani; Vegetarian Food)

is true then this sentence as a whole must be true



CONTD...

- Consider the substitution of a non-vegetarian restaurant such as Ay Caramba's for the variable x.

Vegetarian Restaurant (AyCaramba)

=> Serves (AyCaramba; Vegetarian Food)



CONTD...

Since the antecedent of the implication is False.

Consider the following substitution.

Vegetarian Restaurant(Carburetor))

=> Serves(Carburetor; Vegetarian Food)

Here the antecedent is still false and hence the rule remains true under this kind of irrelevant substitution.



INFERENCE

- The ability to add valid new propositions to a knowledge base.
- Modus ponens is a familiar form of inference that corresponds to what is informally known as if-then reasoning.
- The modus ponens is defined as follows, where α and β should be taken as FOPC formulas.

$$\frac{\alpha \quad \alpha \Rightarrow \beta}{\beta}$$

- schemas like this indicate that the formula below the line can be inferred from the formulas above the line by some form of inference.



CONTD...

- Modus ponens simply states that if the left-hand side of an implication rule is present in the knowledge base, then the right-hand side of the rule can be inferred.
- The left hand side of an implication as the antecedent, and the right-hand side as the consequent.
- consider the following example

$$\frac{\begin{array}{l} \textit{VegetarianRestaurant}(\textit{Rudys}) \\ \forall x \textit{VegetarianRestaurant}(x) \Rightarrow \textit{Serves}(x, \textit{VegetarianFood}) \end{array}}{\textit{Serves}(\textit{Rudys}, \textit{VegetarianFood})}$$

- The formula **Vegetarian Restaurant (Rudys)** matches the antecedent of the rule, thus allowing us to use modus ponens to conclude

Serves(Rudys; Vegetarian Food)



CONTD...

- Modus ponens used practically in two ways
 - Forward Chaining
 - Backward Chaining
- The forward chaining approach has

Advantage

- present in the knowledge base when needed, since in a sense all inference is performed in advance.
- This substantially reduce the time needed to answer subsequent queries.

Disadvantage of this approach

- Facts may be inferred and stored that will never be needed.
- **Production systems**, which are heavily used in cognitive modeling work, are forward chaining inference systems.



CONTD...

- In **backward chaining**, modus ponens is run in reverse to prove specific propositions, called queries.
- The first step is to see if the query formula is true by determining if it is present in the knowledge base.
- If it is not, then the next step is to search for applicable implication rules present in the knowledge base.
- An applicable rule is one where the consequent of the rule matches the query formula.
- If there are any such rules, then the query can be proved - to be true.
- The **Prolog programming language** is a backward chaining system that implements this strategy.

